EXP NO: 3

DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS

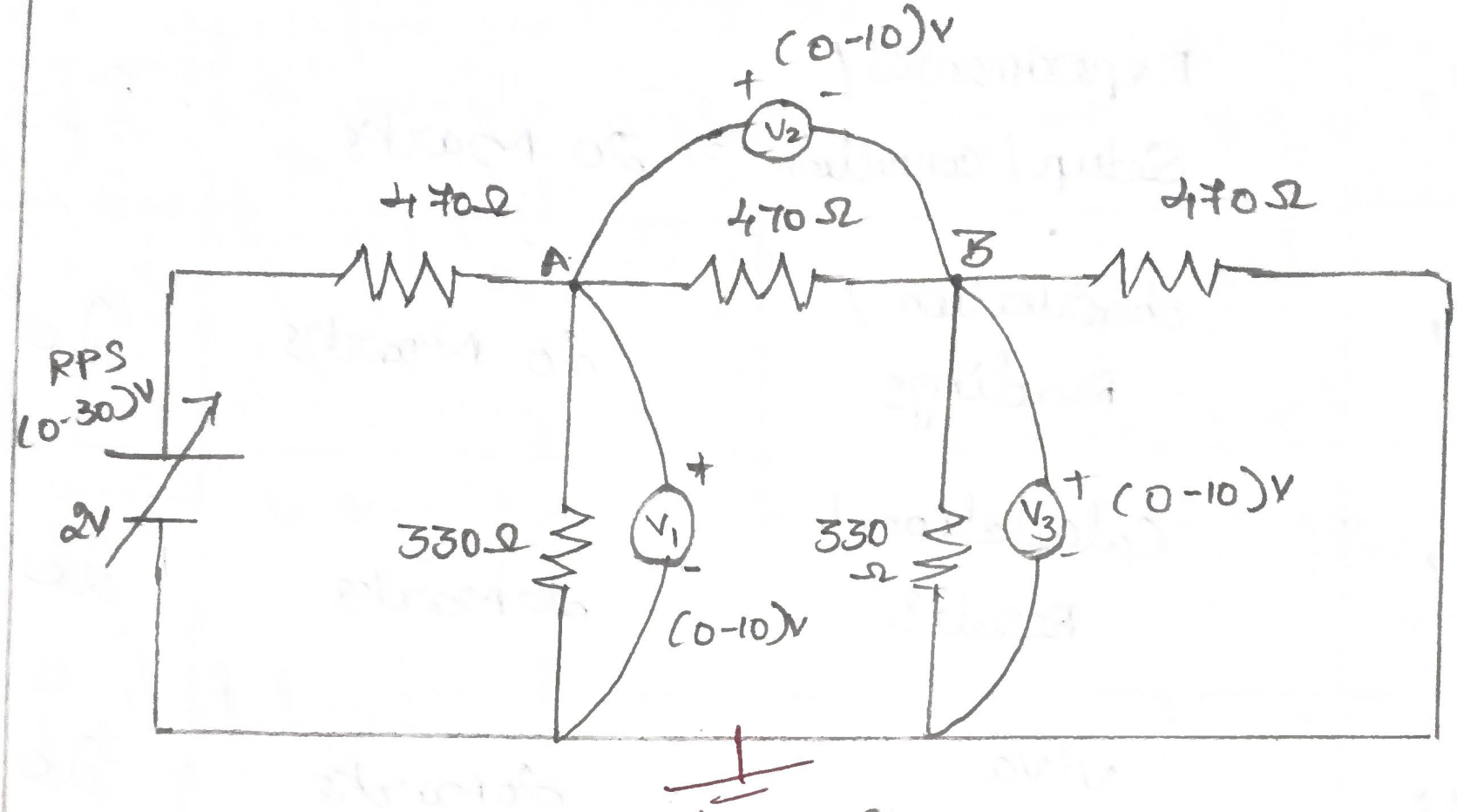
AIM:

To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

Apparatus Required:

SL.NO	Apparatus	Specification	Quantity
1,	Regulated power supply (RPS)	(0-30V).	1
2,	Multimeter	–	1
3,	Resistor	470Ω, 330Ω	3.2
4,	Bread board	–	1

Circuit diagram :



Nodal circuit.

Tabular Column

Parameters	Theoretical	practical
2V	0.92	0.96
5V	2.61	2.68
10V	5.39	5.47
14 V	7.59	7.66

11/2/26 verified

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Calculations:

* for 2V

$V_1 = 0.20$, $V_2 = 0.72$

$V_1 + V_2 = V$

$V = 0.92V$

* for 5V

$V_1 = 0.60$, $V_2 = 2.01$

$V = V_1 + V_2$

$V = 0.60 + 2.01$

$V = 2.61V$

* for 10V

$V_1 = 1.30$; $V_2 = 4.09$

$V = V_1 + V_2$

$V = 5.39V$

PROCEDURE:

1, Give connections as per the circuit diagram

2, Switch ON the supply, vary the RPS and set a particular input voltage.

3, Note down the reading of ammeters and voltmeters and tabulate them.

4, vary the RPS for different input voltages and note down the reading of all the meters.

5, Reduce the RPS to its minimum value and switch off the supply.

6, using the tabulated values, verify Kirchoff's law practically, and verify it theoretically.

Marks split-up :

S.NO	Description	Marks allotted	Mark obtained
1,	Experimental / Setup / connection	20 Marks	20
2,	Observation / Readings	20 Marks	20
3,	Calculations / Results	20 Marks	20
4,	Viva	20 Marks	20
5,	Record	20 Marks	20
		Total :	100

100

11/2/26

RESULT:

Thus, the Nodal analysis is verified practically and theoretically. The resultant voltages for 2 V supply are:

a) The voltage V_1 is 0.96 V.

b) The voltage V_2 is 2.68 V.

c) The voltage V_3 is 5.47 V.

d) The voltage V_4 is 9.66 V